

環境：

1. WWW Server：120.107.152.251 (Ubuntu 22.04)
2. Backup Server：120.107.152.3 (NAS: Network Attached Storage)



(QNAP TS-509Pro，Intel Celeron 1.6GHz CPU, 1 (→2) GB DDRII, 2008)

目標：

定期將 WWW Server 的 Web Pages + Database 備份至 Backup Server

設定：

1. WWW Server

- (a) 於家目錄建立 .netrc 檔 (自動登入 FTP server)

```
machine 120.107.152.3 login xxxxxxxx password *****
```

- (b) 更改 .netrc 的權限

```
$ chmod 600 .netrc
```

- (c) 確定 Backup Sever (120.107.152.3) 是否有 xxxxxxxx 這個 account，且密碼亦如 .netrc 內容

- (d) 資料庫備份設定 (自動登入)

1. \$ cd // 回家目錄
2. \$ mkdir .mysql
3. \$ chmod 700 .mysql // 只讓你自己可以存取這子目錄
4. \$ cd .mysql
5. \$ touch mysqldump.cnf // 建立資料庫自動登入帳密資訊
\$ chmod 600 mysqldump.cnf // 別讓裡面帳密被其他人看到
\$ vi mysqldump.cnf
[mysqldump]
user=s1061000
password=S10612100034\$

- (e) 撰寫 **shell script** 執行自動 FTP 備份 (請注意檔案路徑，**change /home2/s1061000 to your home directory**)

```
#!/bin/bash
# ftp <-- This is comment
cd /home2/s1061000
# Dump your database,
mysqldump --defaults-extra-file=~/.mysql/mysqldump.cnf
--databases s1061000 > s1061000.sql
# Pack your website (www sub-directory) & database dump file
tar zcf s1061000-`date +%d`.tar.gz s1061000.sql www/
ftp -i << backup
open 120.107.152.3
bin
lcd /home2/s1061000
put s1061000-`date +%d`.tar.gz
quit
backup
rm s1061000.sql s1061000-`date +%d`.tar.gz
# Instead of deleting these two files, you may consider
# moving them to some local backup directory so that
# you now have two copies of your backup.
```

- (f) 將 (e) 撰寫 shell script 儲存為 auto_ftp.sh，並將其權限改為 700
(g) 設定 crontab (注意檔案路徑之正確性)

```
$ crontab -e
```

```
0 4 * * 0 /home2/s1061000/auto_ftp.sh
```

Auto_FTP crontab 設定日期 (6/19) 與時間 (與學號關聯，如下)

s1061001	2023/06/19	10:01
s1061002	2023/06/19	10:02
...		
s1061101	2023/06/19	11:01
s1061102	2023/06/19	11:02
...		
s***abcd	2023/06/19	09:cd

<<Basic FTP Commands>>

?	<i>to request help or information about the FTP commands</i>
ascii	<i>to set the mode of file transfer to ASCII (this is the default and transmits seven bits per character)</i>
binary	<i>to set the mode of file transfer to binary (the binary mode transmits all eight bits per byte and thus provides less chance of a transmission error and must be used to transmit files other than ASCII files)</i>
bye	<i>to exit the FTP environment (same as quit)</i>
cd	<i>to change directory on the remote machine</i>
close	<i>to terminate a connection with another computer</i>
delete	<i>to delete (remove) a file in the current remote directory (same as rm in UNIX)</i>
get	<i>to copy one file from the remote machine to the local machine</i>
help	<i>to request a list of all available FTP commands</i>
lcd	<i>to change directory on your local machine (same as UNIX cd)</i>
ls	<i>to list the names of the files in the current remote directory</i>
!ls	<i>to list the names of the files in the current local directory</i>
mkdir	<i>to make a new directory within the current local directory</i>
mget	<i>to copy multiple files from the remote machine to the local machine; you are prompted for a y/n answer before transferring each file</i>
mput	<i>to copy multiple files from the local machine to the remote machine; you are prompted for a y/n answer before transferring each file</i>
open	<i>to open a connection with another computer</i>
put	<i>to copy one file from the local machine to the remote machine</i>
pwd	<i>to find out the pathname of the current directory on the remote machine</i>
quit	<i>to exit the FTP environment (same as bye)</i>
rmdir	<i>to remove (delete) a directory in the current remote directory</i>